**J.H. CERILLES STATE COLLEGE**
**School of Engineering and Technology**
*Mati, San Miguel Zamboanga del Sur*

# EVENT BUDGET MANAGER

A CAPSTONE PROJECT

Presented to the faculty of

School of Engineering and Technology

In Partial Fulfilment

Of the Requirements for the Degree

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

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2024

## APPROVAL SHEET

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## ACKNOWLEDGEMENT

Completing the researchers' capstone project was a journey marked by the unwavering support of numerous individuals, to whom we owe immense gratitude. Their instructor, Mrs. Maribel M. Estioco, played an important role, offering consistent guidance and support from inception to completion. Her feedback and clarifications were instrumental in steering their project towards success.

They sincerely appreciate their esteemed panelists, Edilberta G. Naparan, Marvin C. Mejorada, and Amor Dianne A. Carillo, whose insightful comments and suggestions greatly enhanced the refinement of their work. Their contributions proved invaluable in shaping the final outcome.

Special recognition is also reserved for their capstone adviser, Ms. Mary Dame Angelica M. Canillo, whose unwavering support and guidance were integral throughout the process.

Additionally, they acknowledge the enduring support of their parents and guardians and the guidance of their classmates and friends during challenging moments.

Lastly, they thank the Divine for blessing them with the wisdom and knowledge to complete this project.

**Researchers**

## EXECUTIVE SUMMARY

Celebrating positive occasions fosters happiness through socialization, bonding, and the creation of cherished memories. However, organizing events entails meticulous budgeting, which can pose significant challenges due to various factors such as venue, attire, and catering. This paper addresses the hurdles of budget planning by introducing the Event Budget Manager. Drawing upon insights from "The Importance of Social Connection for Happiness and Well-being" by Barnes and Duck (2006), which highlights the significance of celebrating positive moments in fostering social bonds and positive emotions, this research proposes a solution to improve event budgeting processes.

The Event Budget Manager was designed to facilitate efficient budget planning by providing a platform for resource owners and suppliers to showcase their services and prices. Organizers can easily browse through different offerings, select suitable options, and estimate their budgets swiftly. Moreover, the system enables organizers to provide feedback, fostering communication between organizers and suppliers, thus ensuring alignment with the event's vision.

In conclusion of this study, the developed Event Budget Manager was designed to allow users to have convenient, fast, and easy budgeting for the elements of an event. It provides a feature that allows event organizers to click and select needed resources for the specific event conveniently. The developed system enables suppliers and resource owners to post their resources and prices. This provides them an opportunity to attract more event organizers and increase profits. It is very useful for them since there will be a broader customer base that can see their resources. The system has been tested, and its effectiveness is very useful based on the beta testing conducted. Users provided positive feedback regarding its effectiveness.

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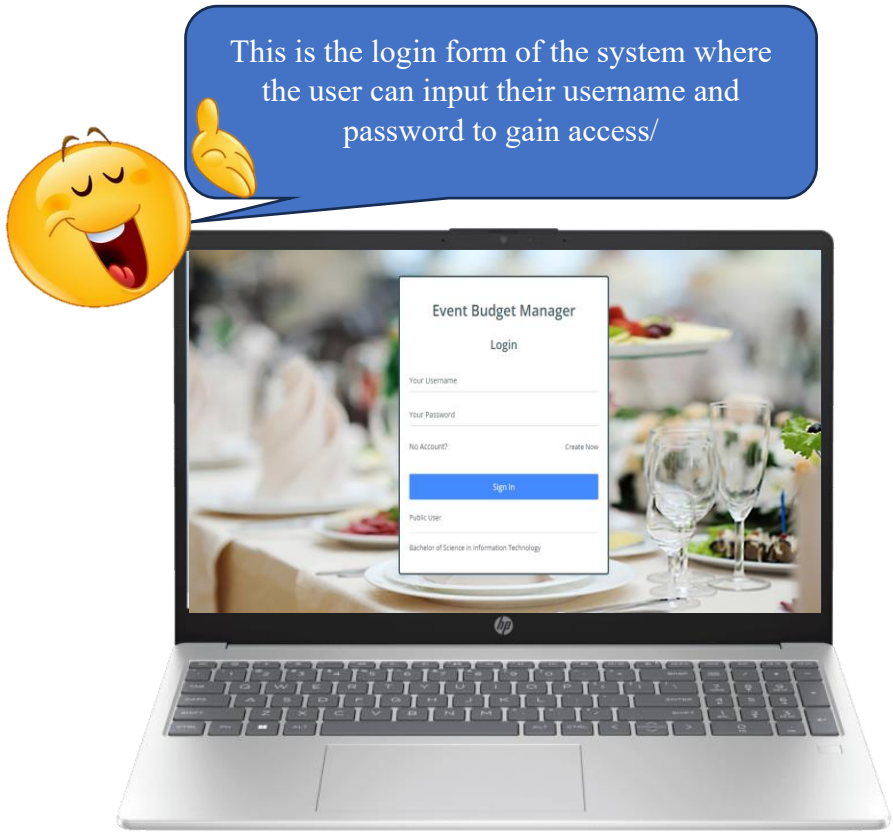
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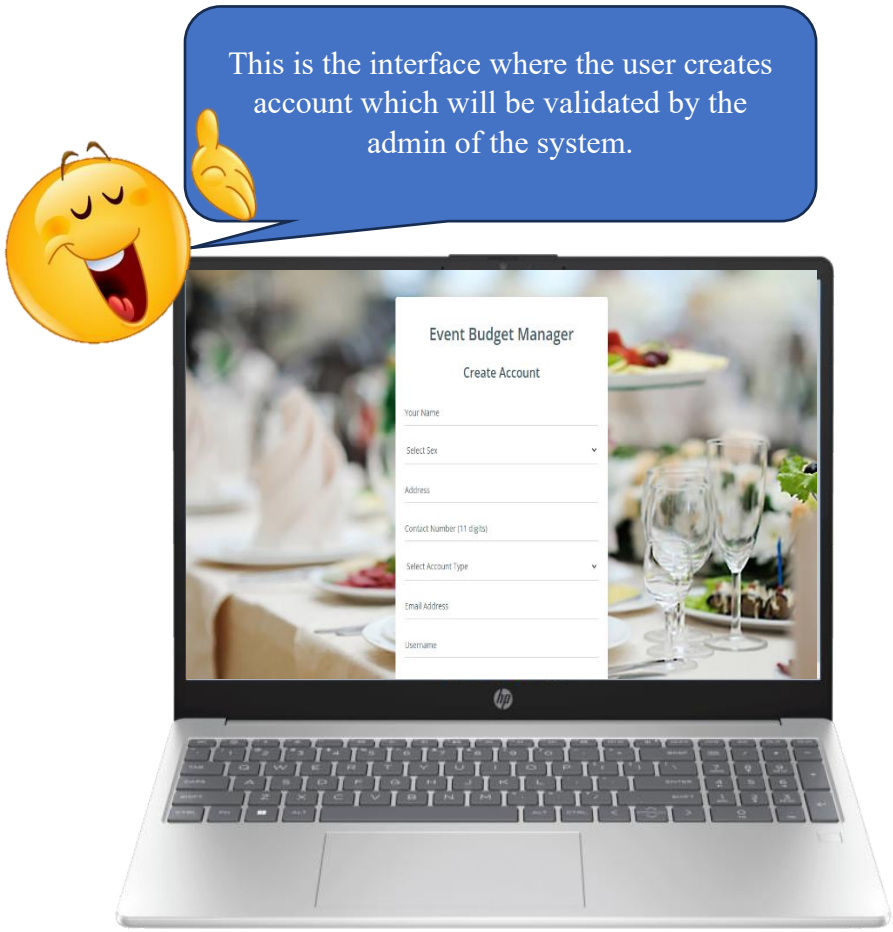
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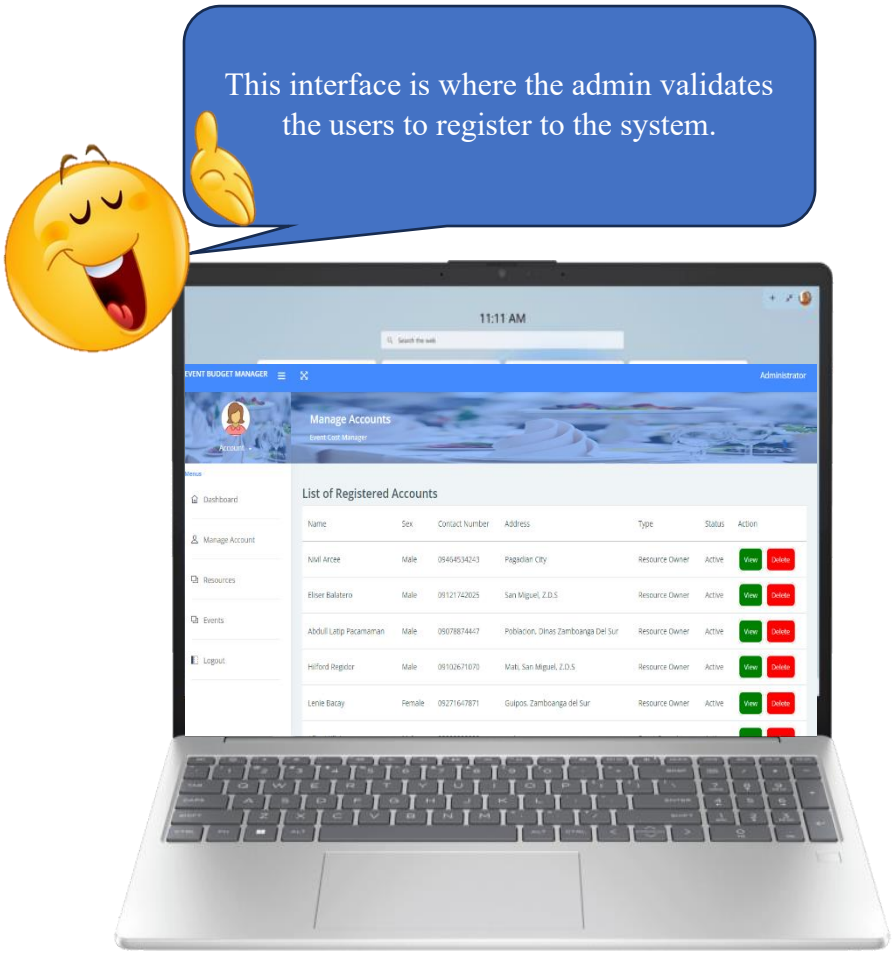
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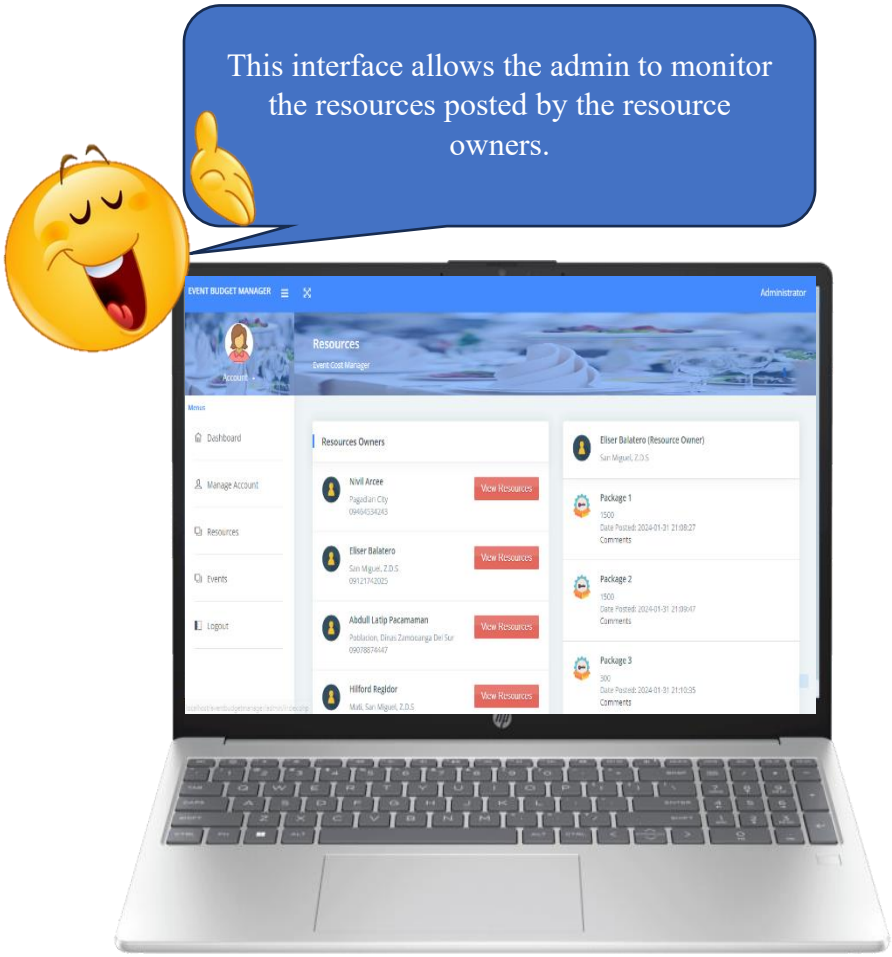
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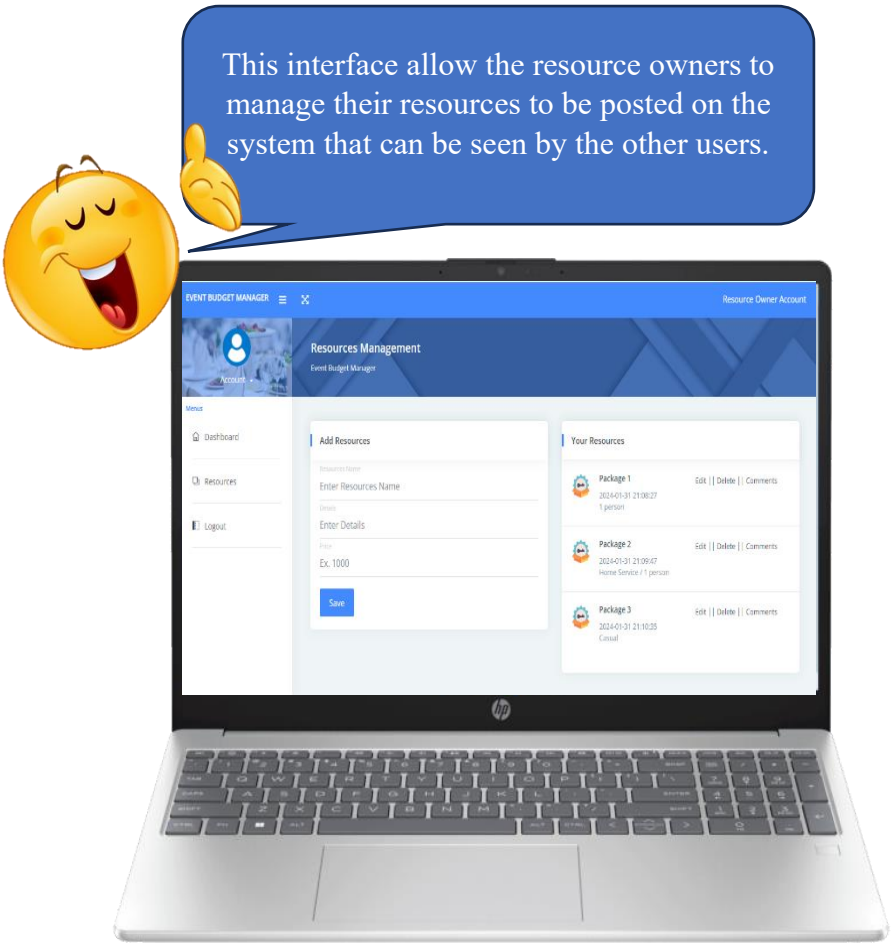
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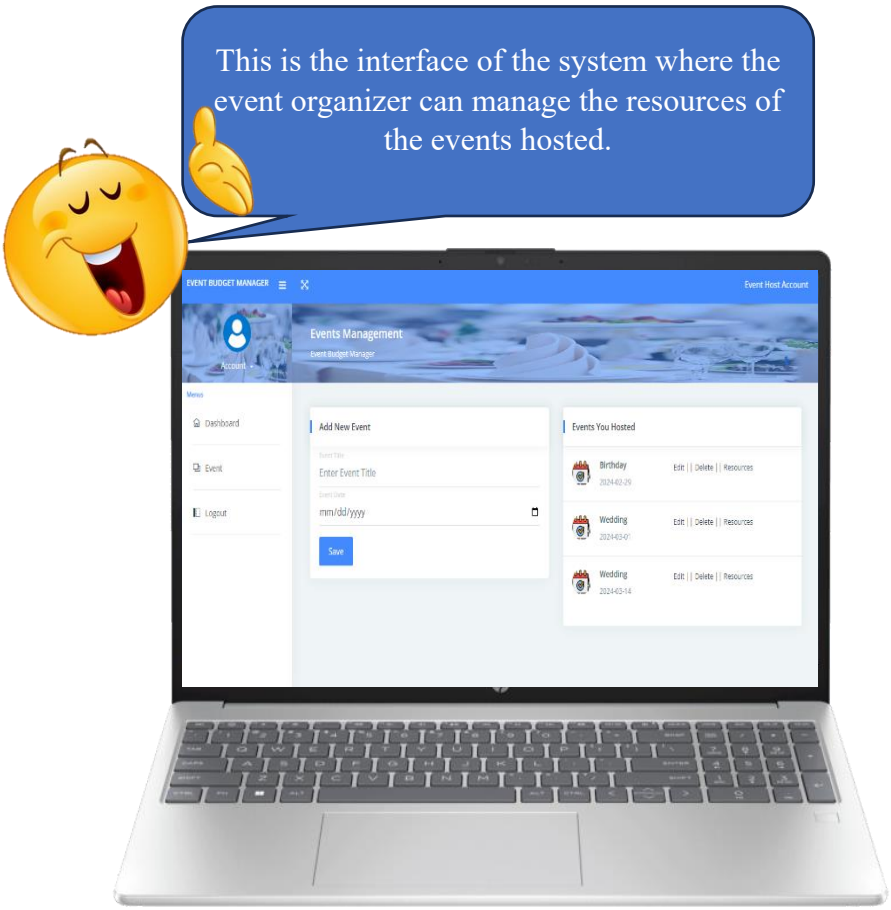


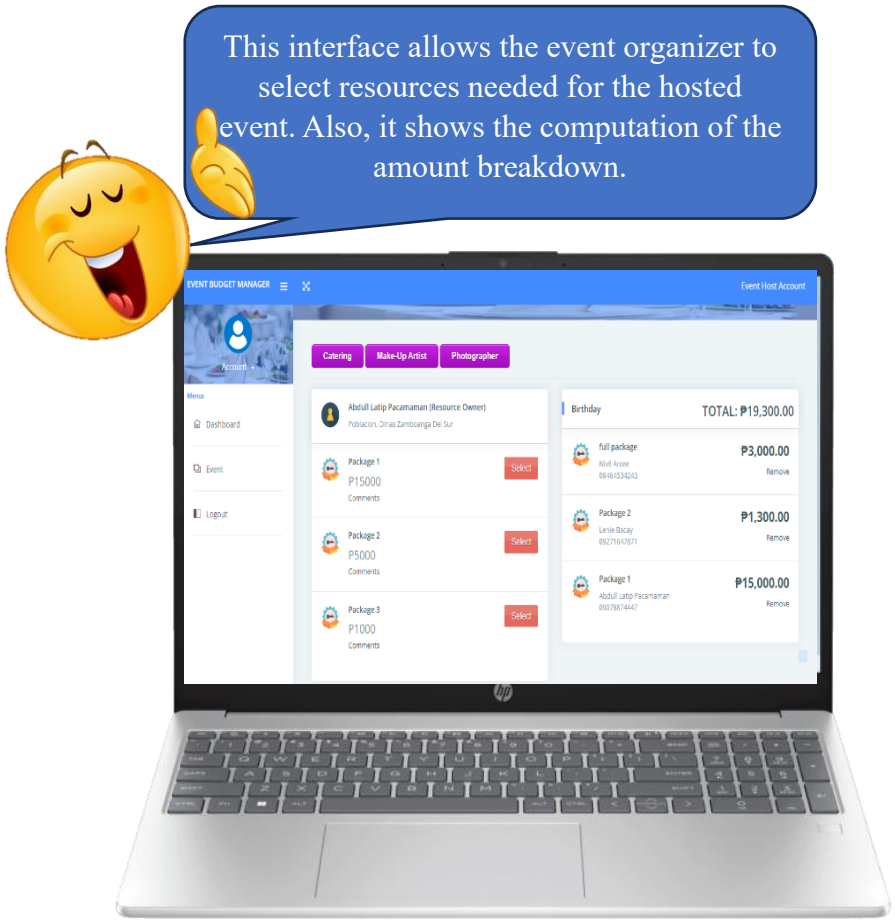



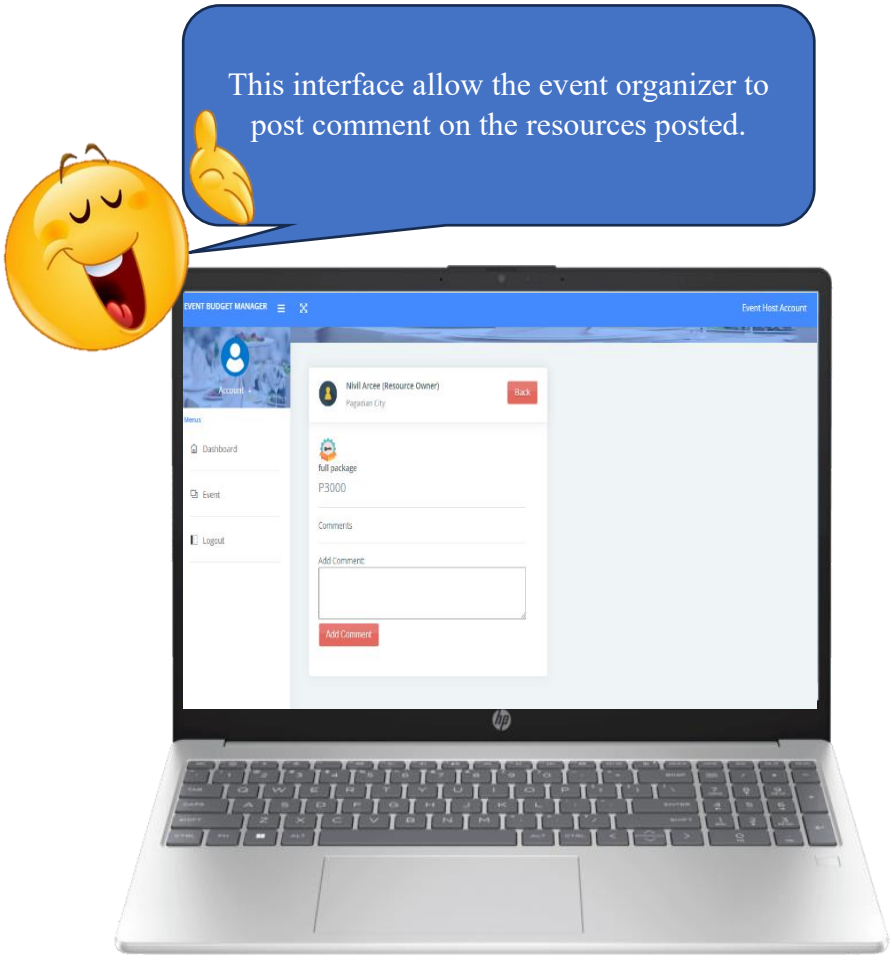

This interface is where the admin validates the users to register to the system.
11:11 AM
EVENT BUDGET MANAGER
Administrator
Manage Accounts
Event Cost Manager
Dashboard
Manage Account
Resources
Events
Logout
List of Registered Accounts
Name
Sex
Contact Number
Address
Type
Status
Action
Nivil Arcee
Male
09464534243
Pagadian City
Resource Owner
Active
Eliser Balatero
Male
09121742025
San Miguel, Z.D.S
Resource Owner
Active
Abdull Latip Pacamaran
Male
09078874447
Poblacion, Dinas Zamboanga Del Sur
Resource Owner
Active
Hilford Regidor
Male
09102671070
Mati, San Miguel, Z.D.S
Resource Owner
Active
Lenie Bacay
Female
09271647871
Gulpos, Zamboanga del Sur
Resource Owner
Active
View
Delete


This interface allows the admin to monitor the resources posted by the resource owners.
EVENT BUDGET MANAGER
Administrator
Resources
Event Cost Manager
Dashboard
Manage Account
Resources
Events
Logout
Resources Owners
Nivil Arcee
Pagadian City
View Resources
Eliser Balatero
San Miguel, Z.D.S
View Resources
Abdull Latip Pacamaran
Poblacion, Dinas Zamoaanga Del Sur
View Resources
Hilford Regidor
Mati, San Miguel, Z.D.S
View Resources
Eliser Balatero (Resource Owner)
San Miguel, Z.D.S
Package 1
1500
Comments
Package 2
1500
Comments
Package 3
300
Comments

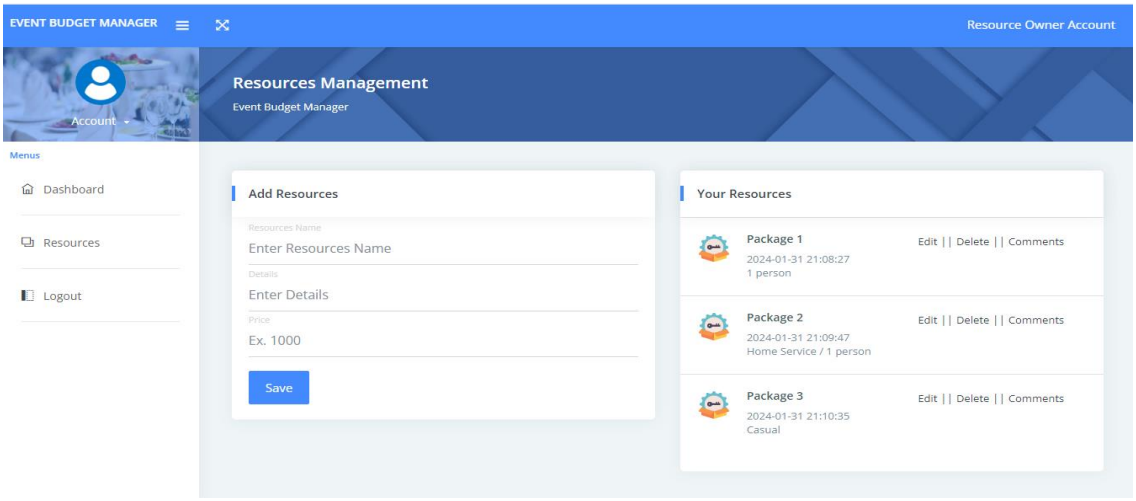

This interface allow the resource owners to manage their resources to be posted on the system that can be seen by the other users.
EVENT BUDGET MANAGER
Resource Owner Account
Resources Management
Event Budget Manager
Dashboard
Resources
Logout
Add Resources
Enter Resources Name
Enter Details
Ex. 1000
Save
Your Resources
Package 1
Package 2
Package 3
Edit | | Delete | | Comments


This is the interface of the system where the event organizer can manage the resources of the events hosted.
EVENT BUDGET MANAGER
Event Host Account
Events Management
Event Budget Manager
Dashboard
Event
Logout
Add New Event
Enter Event Title
mm/dd/yyyy
Save
Events You Hosted
Birthday
Wedding
Wedding
Edit | | Delete | | Resources

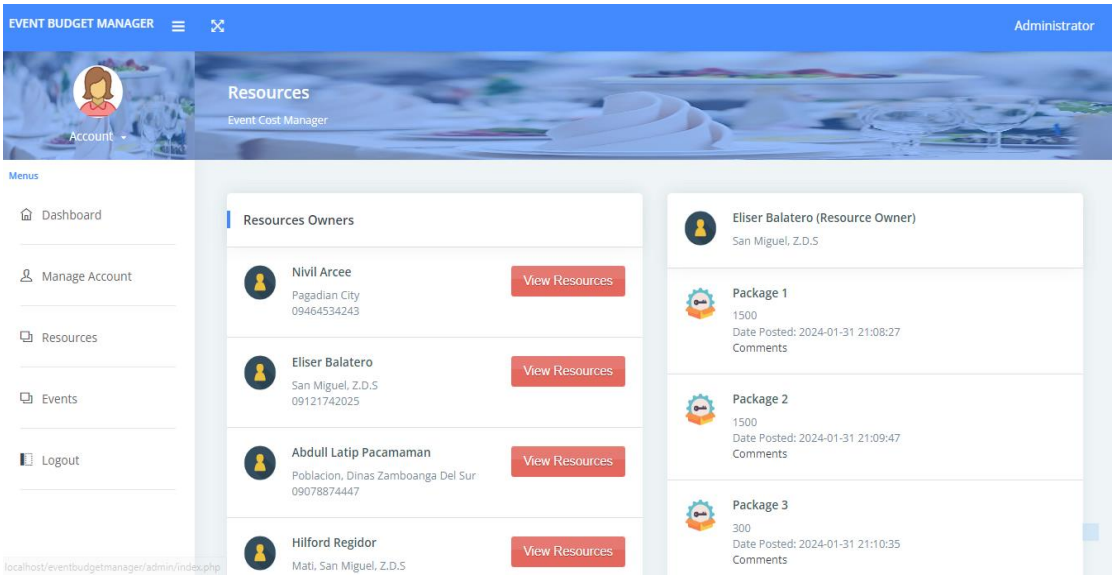

This interface allows the event organizer to select resources needed for the hosted event. Also, it shows the computation of the amount breakdown.
EVENT BUDGET MANAGER
Event Host Account
Catering
Make-Up Artist
Photographer
Abdull Latip Pacamaman (Resource Owner)
Package 1
P15000
Package 2
P5000
Package 3
P1000
Birthday
TOTAL: ₱19,300.00
full package
₱3,000.00
Package 2
₱1,300.00
Package 1
₱15,000.00


This interface allow the event organizer to post comment on the resources posted.
EVENT BUDGET MANAGER
Event Host Account
Nivil Arcee (Resource Owner)
full package
P3000
Comments
Add Comment:
Add Comment

## B. Sample Input/Output




The image above is the sample input of the system where the supplier fill-in the form with regards to the services offered. This information will be stored in the system database.




The image above is the sample output of the system. Wherein the information that was stored in the database was retrieved. It allows the event organizer to view the stored services by different suppliers.

## C. Photo Documentation







The images above are some of the pictures captured during the process in the journey of this study. It shows from the interview up to the testing of the system.

## D. Curriculum Vitae



**Personal Data**

Name : Margeline B.Baco

Gender : Female

Data of Birth : July, 08, 1998

Place of Birth : Barangay Old Mirapao, Dinas, Zamboanga Del Sur

Civil Status : Single

Citizenship : Filipino

Religion : Iglesia Sa Dios Spirit Santo

Address : Barangay Old Mirapao, Dinas, Zamboanga Del Sur

Father's Name : Marvin Baco

Occupation : Farmer

Present Address : Barangay Old Mirapao, Dinas, Zamboanga Del Sur

Mother's Name : Melinda Baco

Occupation : Housekeeper

Present Address : Barangay Old Mirapao, Dinas, Zamboanga Del Sur

**EDUCATIONAL BACKGROUND**

| Education | School | Year |
|---|---|---|
| Primary | Pasig City Elementary School | 2011-2012 |
| Secondary | Old Mirapao National High School | 2016-2017 |
| Tertiary | J.H. Cerilles State College | 2023-2024 |
| Degree | Bachelor of Science in Information Technology | |

**Curriculum Vitae**



**Personal Data**

Name : Alexandrea Nicole B. Paña

Gender : Female

Data of Birth : November 3, 1998

Place of Birth : Barangay Bulawan, Lapuyan, Zamboanga del Sur

Civil Status : Single

Citizenship : Filipino

Religion : Catholic

Address : Barangay Bulawan, Lapuyan, Zamboanga del Sur

Mother's Name : Nenita Paña

Occupation : Housekeeper

Present Address : Barangay Bulawan, Lapuyan, Zamboanga del Sur

**EDUCATIONAL BACKGROUND**

| Education | School | Year |
|---|---|---|
| Primary | Bulawan Elementary School | 2011-2012 |
| Secondary | Betinan National High School | 2017-2018 |
| Tertiary | J.H. Cerilles State College | 2023-2024 |
| Degree | Bachelor of Science in Information Technology | |

**Curriculum Vitae**



**Personal Data**

Name : Danimar Y. Sedenio

Gender : Female

Data of Birth : August 24, 2000

Place of Birth : Panubigan, Pitogo, Zamboanga Del Sur

Civil Status : Single

Citizenship : Filipino

Religion : Catholic

Address : Panubigan, Pitogo, Zamboanga Del Sur

Father's Name : Danilo G. Sedenio Sr.

Occupation : Fishermen

Present Address : Panubigan, Pitogo, Zamboanga Del Sur

Mother's Name : Margie Y. Sedenio

Occupation : Housekeeper

Present Address : Panubigan, Pitogo, Zamboanga Del Sur

**EDUCATIONAL BACKGROUND**

| Education | School | Year |
|---|---|---|
| Primary | Panubigan Elementary School | 2012-2013 |
| Secondary | Panubigan National High School | 2016-2017 |
| Tertiary | J.H. Cerilles State College | 2023-2024 |
| Degree | Bachelor of Science in Information Technology | |

**Curriculum Vitae**



**Personal Data**

| | |
|---|---|
| Name | : Anamaria Sebilo Sumarago |
| Gender | : Female |
| Data of Birth | : April 18,2000 |
| Place of Birth | : Provincial Hospital Pagadian City |
| Civil Status | : Single |
| Citizenship | : Filipino |
| Religion | : Church of God |
| Address | : Purok Acasia, Bulawan, San Miguel, Zamboanga Del Sur |
| Father's Name | : Anatalio M. Sumarago |
| Occupation | : Farmer |
| Present Address | : Purok Acasia, Bulawan, San Miguel, Zamboanga Del Sur |
| Mother's Name | : Maria Teresa Sebilo Sumarago |
| Occupation | : Deceased |
| Present Address | : Deceased |

**EDUCATIONAL BACKGROUND**

| Education | School | Year |
|---|---|---|
| Primary | Bulawan Elementary School | 2012-2013 |
| Secondary | Betinan National High School | 2018-2019 |
| Tertiary | J.H. Cerilles State College | 2023-2024 |
| Degree | Bachelor of Science in Information Technology | |

# CHAPTER I

# BACKGROUND OF THE STUDY

## Introduction

In this generation, where technology plays a central role in assisting people with their everyday proceedings, it is highly regarded as one of the best tools for various transactions and operations. Thanks to its advantages, people are adapting to its capabilities.

Moreover, technology has been found to solve different issues and problems. Through its features, users are being assisted with more opportunities regarding business operations and event management.

In this study, the proponents will develop a system entitled "Event Budget Manager" aiming to provide a solution to the problems encountered. This system offers different features for event organizers to manage event resources more easily and quickly.

## Project Context

Celebrating positive occasions typically brings individuals happiness as they offer opportunities for socialization, bonding, and the creation of unforgettable memories with loved ones. According to the study titled "The Importance of Social Connection for Happiness and Well-being" by Barnes and Duck (2006), celebrating positive moments brings people together to share joy, laughter, and positive emotions, especially in social gatherings, celebrations, and parties. However, budgeting becomes a challenge in this process, considering various elements such as venue, attire, makeup artist, photographer, resource owner, and catering. Managing the budget presents a significant challenge and can be a real headache for the organizer, as computing and comparing costs can be demanding and often lead to budgeting failures. Despite these difficulties, effective budget planning is essential to ensure the event's success and financial stability.

To address the challenges of budget planning for events the proponents developed the Event Budget Manager. This platform would enable suppliers to showcase their packages and prices, allowing organizers to select the needed services and estimate their budget easily. By creating Event Budget Manager, the organizer can browse different price offers and packages and select what suits the budget, the system provides a faster budgeting process. Also, allowing the event organizer to comment about the resources enables suppliers to be aware of the organizers' ideas.

By implementing this system, event organizers will be fed with the necessary information to budget ahead of time effectively. The Event Budget Manager allows the organizer to have fast budgeting and determine the cost of the event. Organizers can access all the relevant information, compare prices and packages, and make informed decisions, all in one place. These eliminate the hassle of manual calculations and extensive research, allowing organizers to plan their budgets efficiently and accurately. Ultimately, the Event Budget manager is an indispensable tool for organizers, ensuring smooth and hassle-free event budgeting.

**Purpose and Description**

The purpose of this study is to provide an Event Budget Manager that will help organizers in planning an event, especially on regarding budgeting, also allowing the event organizer to comment on the packages posted. The study will empower proponents to craft frameworks and solutions aimed at addressing users' concerns effectively. The system also provides privilege to service sellers such as coordinator, venue, attire, make-up artist, designer, photographers, resources owner, and catering. The Event Budget Manager will select entities for the event with their packages and corresponding prices fast and easy.

**General Objectives**

The goal of this study is to develop an Event Budget Manager to provide a solution. It is also adapting the usability of technology and aiming to provide services that will satisfy the users to the problem encountered by the event organizer. Additionally, it adapts the usability of technology and strives to deliver services that meet users' needs.

**Specific Objectives**

- To design a system that will allow the organizer to have convenient, fast, and easy budgeting for the elements of an event.
- To develop a system that allows service sellers to post their packages and prices.
- To test the effectiveness of Event Budget Manager in different event planning.

**Scope and Limitation**

The scope of the study includes the development of Event Budget Manager, it focuses on providing a fast centralized platform where suppliers can showcase their packages and prices, enabling organizers to select the services they need easily, and the system will compute the cost. Providing fast and easy process of budgeting event

resources. Additionally, the event organizer can add comments on the packages. Also, providing a feature that enable the public users to view the packages posted by the suppliers.

The system does not involve the planning of activities regarding the event. Moreover, the system is not accessible without an internet connection. Also, the system does not support uploading pictures of the packages.

**Definition of Terms**

*Event Budget Manager*. A system that allows organizer to select prices and packages needed for the event and what suits the budget.

*Organizer*. The main user of the system that navigates system features for planning an event.

*Resources Provider.* These users will be posting their package services like photography, artist, facilities other needed resources for an event.

*Public User.* These users can view the packages posted by the suppliers.

# CHAPTER II

# REVIEW OF RELATED LITERATURE

This chapter presents studies related to this study. The foreign studies include Automated House Budget Plan Application, Development of Financial Planner Application Software Based on Waterfall Model and Research on Comprehensive Budget Management of Logistics Enterprises. The local studies include Equipment Monitoring and Budget Allocation, Accounting Management System, and Mobile Bookkeeper: Personal Financial Management Application with Receipt Scanner Using Optical Character Recognition.

## Foreign

*Automated House Budget Plan Application*

The system developed by Rizqi et al., 2022 entitled Automated House Budget Plan Application, published in Yogyakarta, Indonesia, is an automated house budget plan used by architects to support their productivity while building a house. This application provides information on material costs along with the budget plan. It has a template that allows architects to choose the resources needed for the house and wages for the workers. The manual creation of a house budget plan by creating a worksheet and filling up information manually, which causes much time and is also prone to human error, will be transmitted electronically. Allowing the architects to create a house budget plan fast and easy with just a few taps, selecting a template, and the suggestions of materials will be shown, wherein the architect will just select and customize needed materials. With this feature, the 2-hour plan used during planning will be lessened to 30 minutes without the possibility of human error because it is computerized. Also, the system has a database where the data will be stored, filtered, and organized, the request for information will be faster, and it is secured.

The study provides a fast budgeting for house planning that architects can use. This application promotes fast planning with just a few taps. With these features, the proponent of Event Budget Manager included this in their study, providing a feature for the organizers to have fast budgeting of resources for a certain event with just a few taps.

*Development of Financial Planner Application Software Based on Waterfall Model*

The study “Development of Financial Planner Application Software Based on Waterfall Model” by Immanuel et al., 2022 in Bandung, Indonesia, addresses the challenges of financial planning, especially for people who are beginning to save

money. It involves careful planning, organizing, managing, and monitoring financial activities. The application assists all users in managing their finances based on their monthly budget. This app will be a great help for the users to control the flow of their money according to their preferred budgets; because of this, they can save and manage their finances efficiently.

The application named Development of Financial Planner Application Software Based on Waterfall Model enables users to manage their finances effectively through a user-friendly monthly budgeting feature. Monthly expense budgeting is done electronically, improving the financial management process. This feature has also been integrated into the Event Budget Manager, basing the developed Financial Planner. Therefore, the Event Budget Manager will enable the users to budget the resources needed for an event. This is a big help to the event organizers in planning and managing event resources. The system allows the organizer to select resources from a wide owner, easing the selection and computation of event budgets.

*Research on Comprehensive Budget Management of Logistics Enterprises*

The study entitled "Research on Comprehensive Budget Management of Logistics Enterprises" created in China, states that the importance of budget management systems developed by Gou et al., 2021. Where employees and stakeholders will monitor, and able to provide allocations on the logistics. Having a system that can provide information in relation to budgeting provides a greater impact and helps to improve overall financial management. In general, having a budget management system enables users to monitor, allocation and have accurate budgeting.

The feature that they found provides importance of having management systems. Where employees can have easy navigation in budgeting the logistics. Providing a system that helps budgeting and improves overall financial management. This concept was being carried out and used by the proponent of the Event Budget Manager as they continued their study.

**Local**

*Equipment Monitoring and Budget Allocation*

According to Inettutor team, 2019, Equipment Monitoring and Budget Allocation developed in the Philippines is a web-based application that allows users to manage equipment inventory, providing accurate information on the status of equipment. Additionally, the system has a feature that allow users to allocate budgets on the equipment, and it can be used for budget planning. Also, the system calculates

the cost of the maintenance of the equipment automatically. Through this system, every business unit's equipment will be allocated with budgets quickly and easily. This system is web-based; therefore, it can be accessed with the internet, anywhere, and on any computer as long as there is an internet connection. The list of business units can be viewed in the system wherein the user will just select the unit and what equipment to allocate with budgets; with just a few taps and simple navigation, the work will be done easily. Another feature of the system is that it contains inventory, this will allow users to manage the equipment tracking; the quantity of working and damaged, dates of the equipment will be thoroughly monitored.

The Equipment Monitoring and Budget Allocation is a web application with a feature allowing users to allocate budgets on the equipment, which is effective for budget planning. Henceforth, the proponent of Event Budget Manager includes the said feature on their system for more effective event budgeting about the resources needed by the organizers.

*Accounting Software in the Philippines*

An application named “Accounting Software in the Philippines”. This application enables Filipinos to manage financial aspects like billings and expenses. The user can be able to view all transactions in relation to the expenses and budgets. Through this app, budgeting and monitoring made easier for Filipinos. The system promotes functionality that allows user to organize the accounting aspects and start effective budgeting (Oojeema team, 2023).

On the system named Accounting Software in the Philippines, it allows users to easily budget financial resources. This feature also became the main core of the system entitled Event Budget Manager, as it focuses on budget allocation, easing the process of budgeting done by organizers. Also, it promotes a more effective monitoring of budgets, allowing organizers decisions with computation automation.

*Mobile Bookkeeper: Personal Financial Management Application with Receipt Scanner Using Optical Character Recognition*

An application developed by Garcia et al., 2021 entitled “Mobile Bookkeeper: Personal Financial Management Application with Receipt Scanner Using Optical Character Recognition” from Manila, Philippines provides a feature where users can manage personal finances. Allowing the user to budget expenses throughout a certain month or year. This application promotes innovations in recording, tracking, and monitoring of finances and budgets of the users. Moreover, the application promotes

the effectiveness of mobile finance apps that assist users in relation to budget management.

The Personal Financial Management App allows users to plan their spending for a particular month or year. This feature helps users manage their expenses more effectively. It was also one of the main reasons for developing the Event Budget Manager. The feature helped the developers to design and test the Event Budget Manager's effectiveness. The said system will enable organizers to budget event resources in a more efficient and fast way.

Based on the abovementioned study, the proponents gathered valuable insights into the methodologies, techniques, and functions applied to existing event budget management systems by thoroughly examining these studies. This knowledge will be instrumental in guiding the development of the Event Budget Manager, ensuring that it incorporates the most effective and up-to-date practices in the field. Moreover, by understanding the strengths and limitations of existing systems, the proponents can identify areas for improvement and innovation, ultimately creating a more effective and user-friendly event budget management tool.

# CHAPTER III

# DESIGN AND METHODOLOGY

This chapter contains the design and methodology used in this study. These research components serve as a guide and reference for the researchers throughout the development of Event Budget Manager.

## Requirement Analysis

The 4D model is a widely used and practical system development framework. It is a structured and iterative approach that helps to ensure that the system meets the needs of the users and is developed in a timely and cost-effective manner (Thiagarajan, et al., 1974).

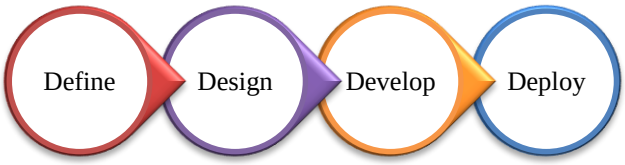



**Figure 1.0 4D Model**

Figure 1.0 4D Model shows the methodology used in this study. It has four phases which are defined, design, develop and deploy. It phases have its own assigned activities that the proponents performed.

## Define Phase

The define phase consists of the identification of the problem through engaging with the current situation of the suppliers and event organizers. Also, in this phase, the proponent determines the solution based on the problem found during identification.

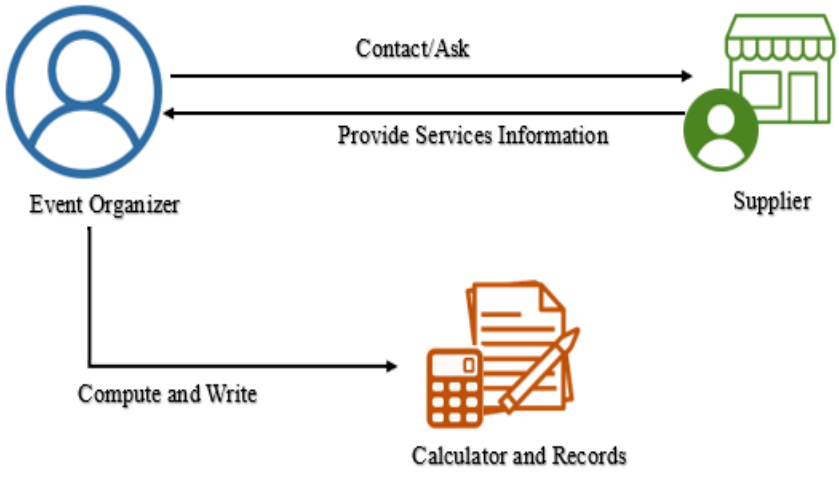



**Figure 2.0 Current Technical Situation**

Figure 2.0 Current Technical Situation shows the scenario of the event organizer and suppliers. The event organizer will contact/ask the resource owners regarding their

services offered and the prices, and the resource owners will provide the information. After collecting details from the resource owners, the event organizer will write the details and compute the cost manually.



**Image 1.0 Data Gathering**

Image 1 shows the data gathering documentation during the collection of data conducted by the proponents. Together with the tourism in-charge which also an event organizer Francis Dohaylongsod, and resource owner make-up artist Dexter Mamadod. Also, some of the resource owners named Antonieta Tompong with her office mates.

**Design Phase**

The design phase consists of the creation of the system diagrams, structure, and other components used for the system design. Also, this is where proponents create the user interface following the buttons, and menus in the diagrams created.

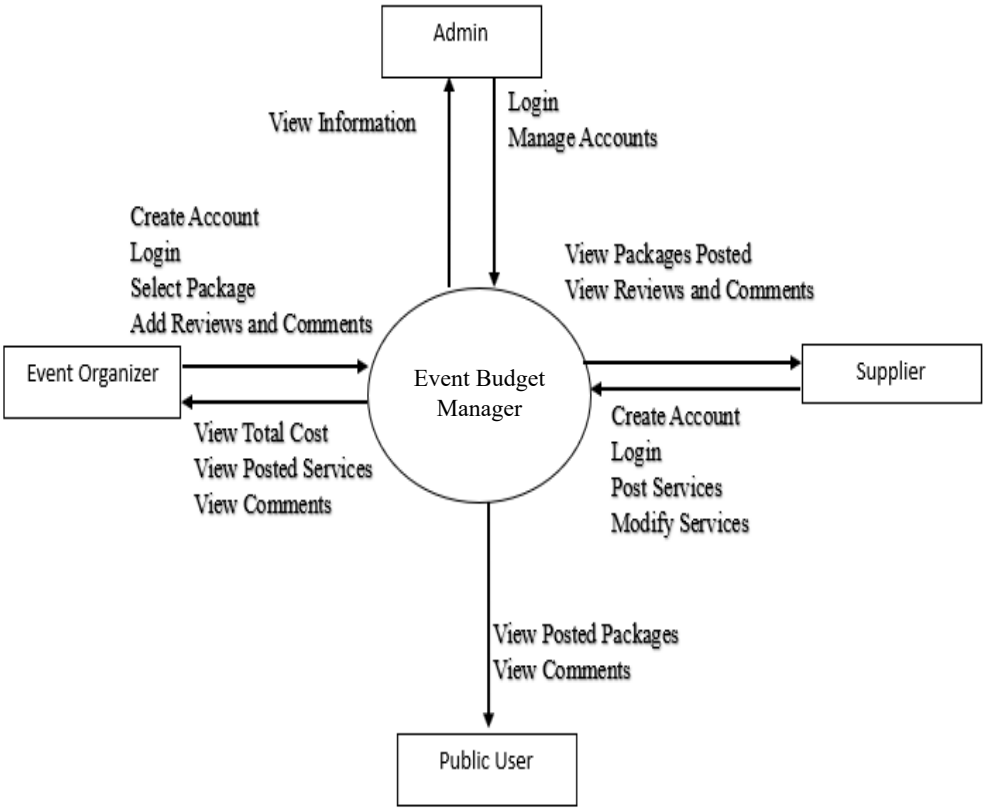



**Figure 3.0 Context Diagram**

Figure 3.0 Context Diagram shows the interaction of the users and the system. The admin can login to the system for user credential verification, and control user's access. The system response to the admin by allowing it to view information within the system. The suppliers can create account, login, and able to post and modify services.

The system response to the resource owners by allowing them to view services posted, and comments. The event organizer can create an account then login, after it, they can select services/packages from the postings of suppliers, also they can add comments to the posted services. The system response by allowing the event organier to view total cost that will be computed automatedly, view posted services, comments. Lastly, the public user can view posted packages and comments.

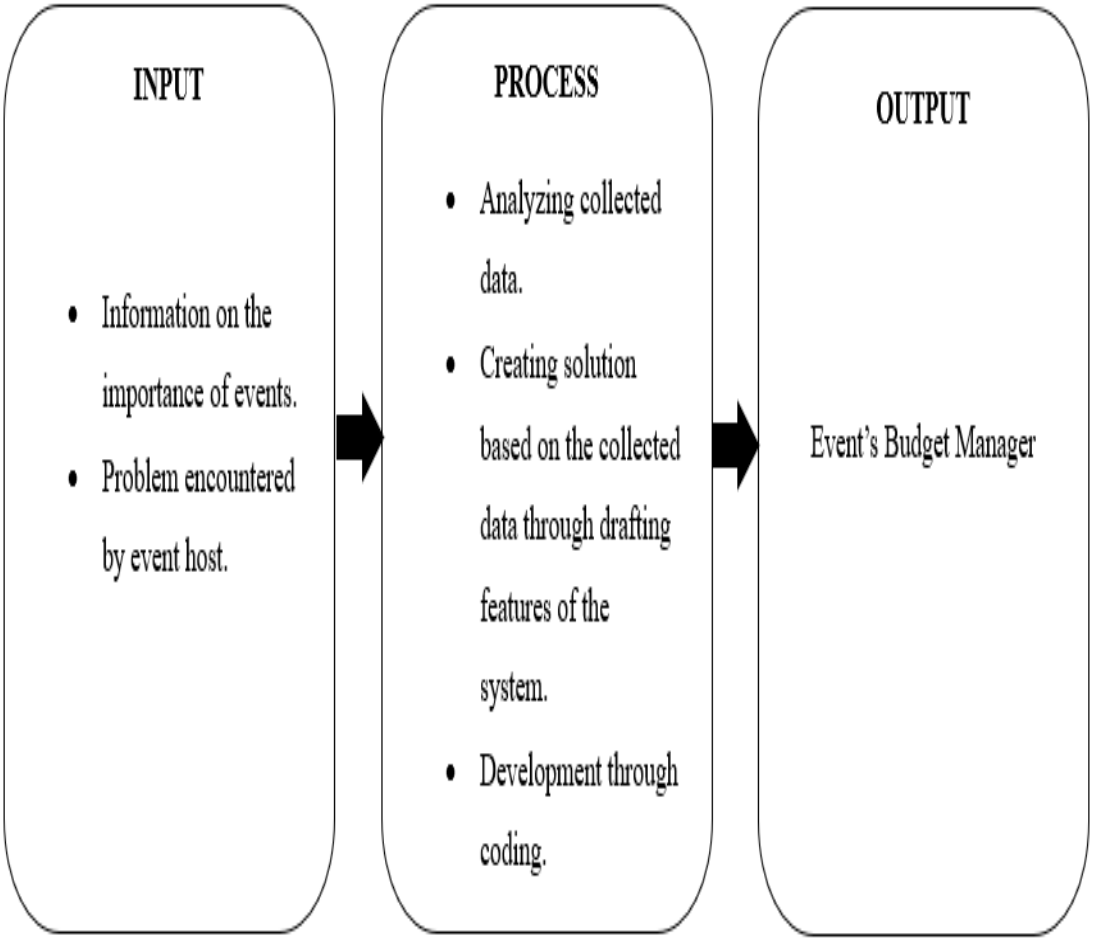



**Figure 4.0 Conceptual Framework (IPO)**

Figure 4.0 Conceptual Framework (IPO) shows the Input, Process and Output of this study. In the outputs, it contains the information about the importance and events and the problem encountered by the event organizer. During the process, the proponents analyze the collected data, create solution data drafting the features of the system, and the development of the system. Lastly, is the output of the study which is the Event Budget Manager System.

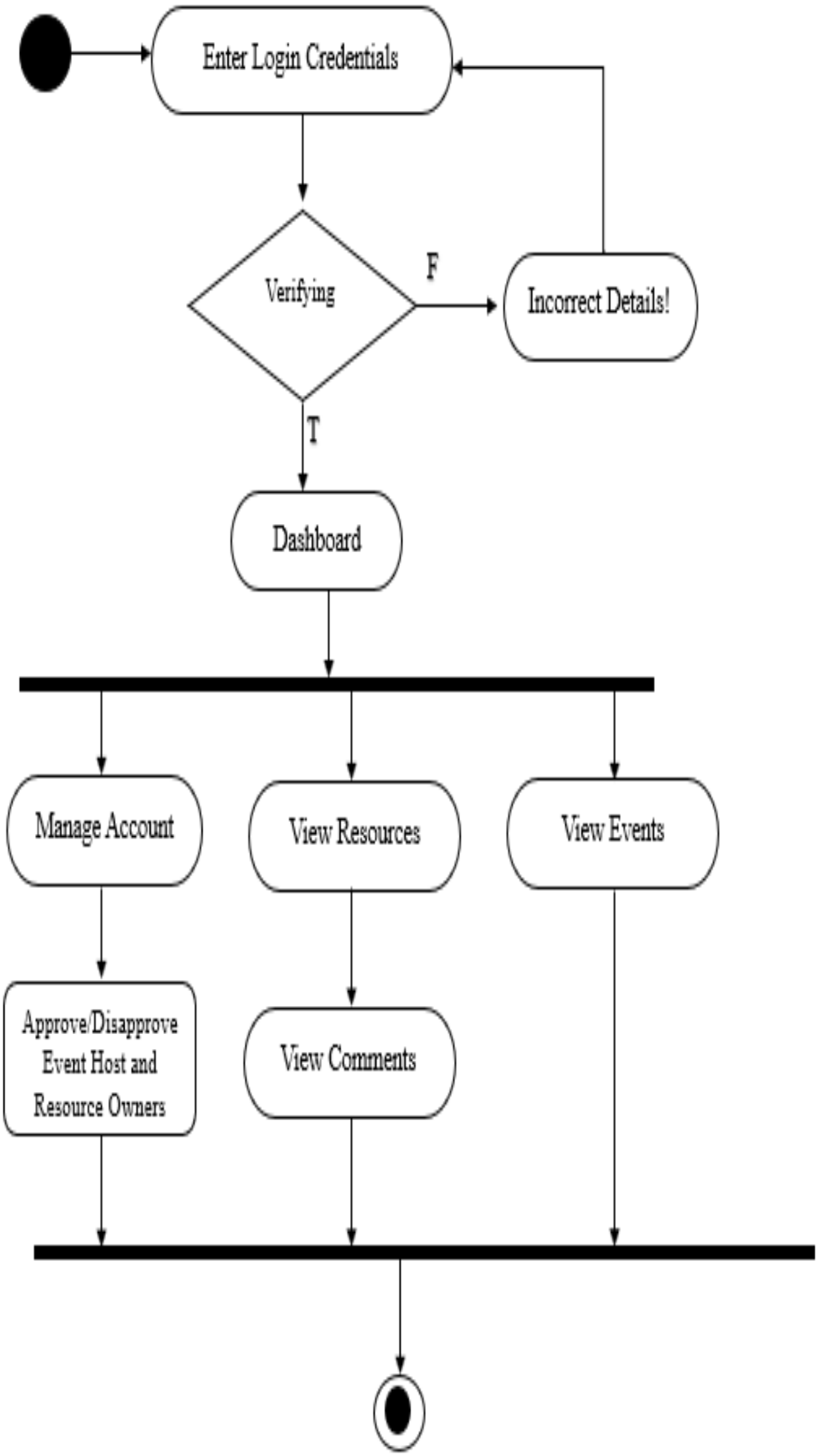



**Figure 5.0 Activity Diagram for Admin**

Figure 5.0 Activity Diagram for Admin shows the activities of the admin when using the system. First, the admin must login to the system, once verified, then proceed to the system dashboard. In the dashboard, the admin can manage accounts, then approve/disapprove event organizer supplier's account. Then able to view resources, view comments and view events information.

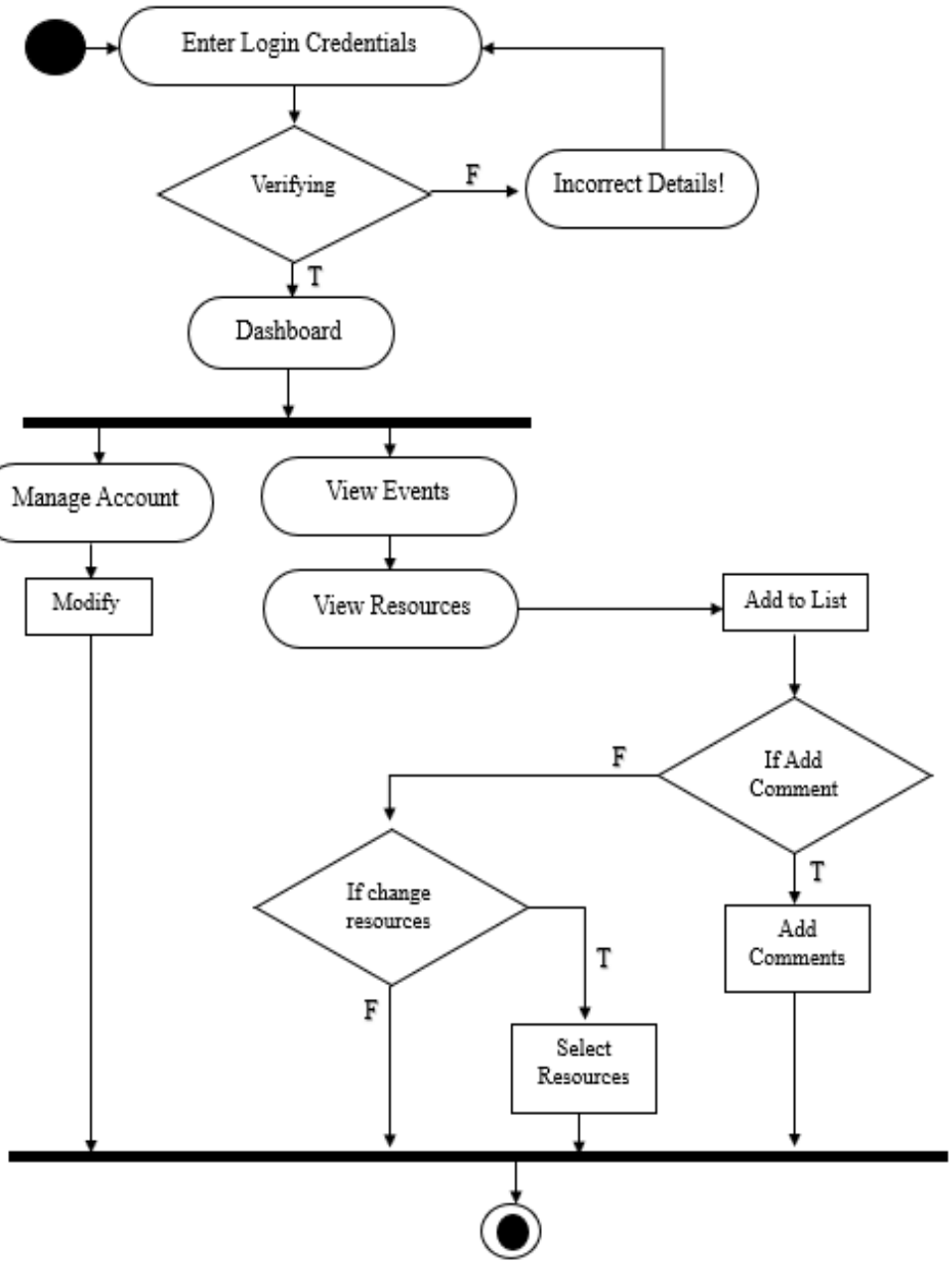



**Figure 6.0 Activity Diagram for Event Organizer**

Figure 6.0 Activity Diagram for Event Organizer shows the activities of the event organizer user when using the system. To start, the event organizer must enter login credentials so that the system can verify its access. Once logged on, the event organizer can view the dashboard and able to choose between the three options which are: manage account, view resources and view event. Under the manage account, there is a button modify so that the event organizer can change details. Under the view resources, the event organizer can add to list, once the button is clicked, the selected resources will be added to the list of and in addition to the calculation. Next, there is a condition “if add comment”, if true then add comments, if false then encounter a new condition “if change resources”, if true, then the event organizer can change the resources and select and new, if false, the event organizer can confirm the list of resources chosen.

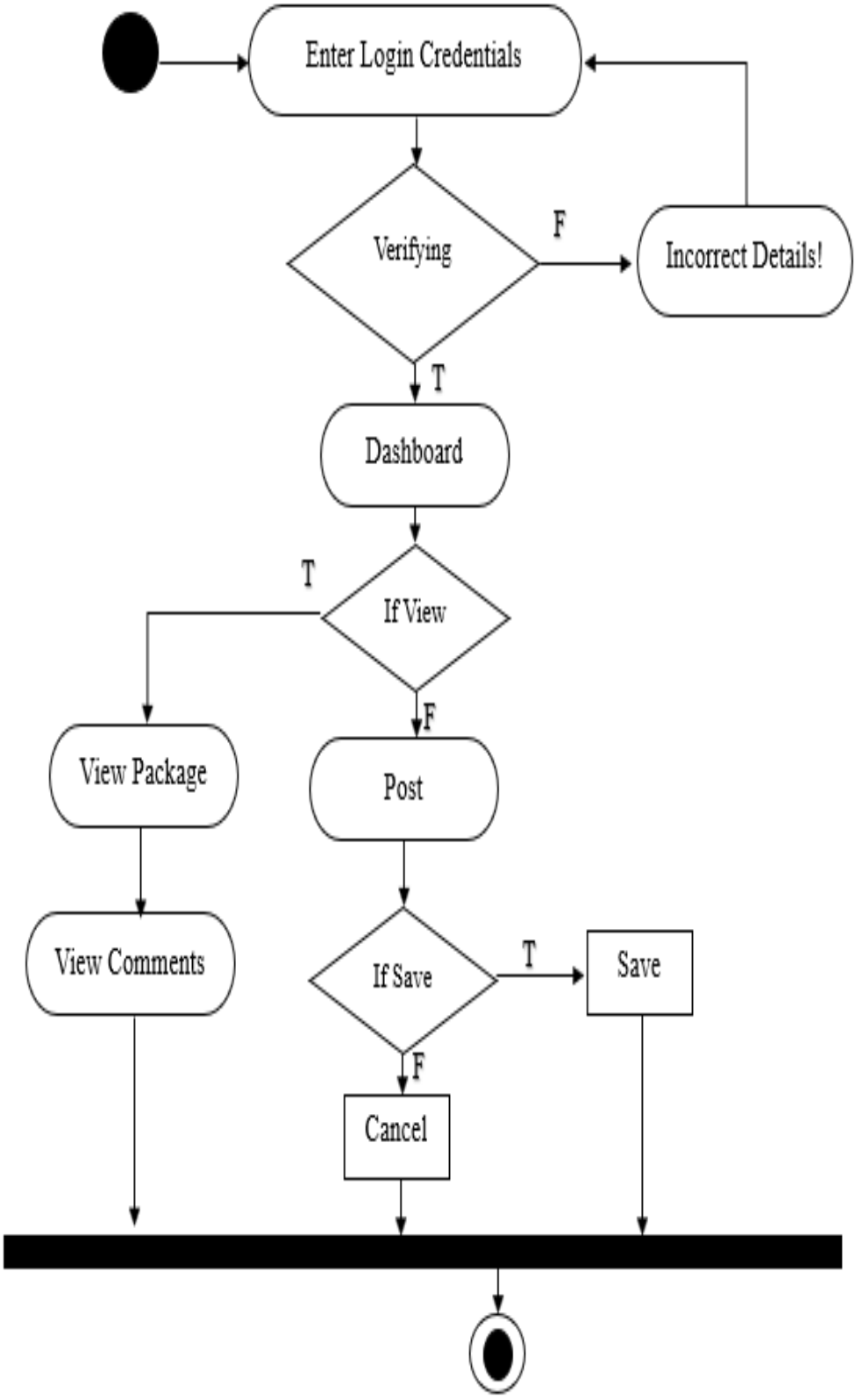



**Figure 7.0 Activity Diagram for Suppliers**

Figure 7.0 Activity Diagram for Suppliers shows the activity of suppliers when accessing the system. The suppliers will be required to enter login credentials for access verification. Once logged on, then able to proceed for the dashboard and post their services offered. Under posting, there is a condition that states if save, if the response is false, then the saving will be canceled, and once true, then the saving will continue.

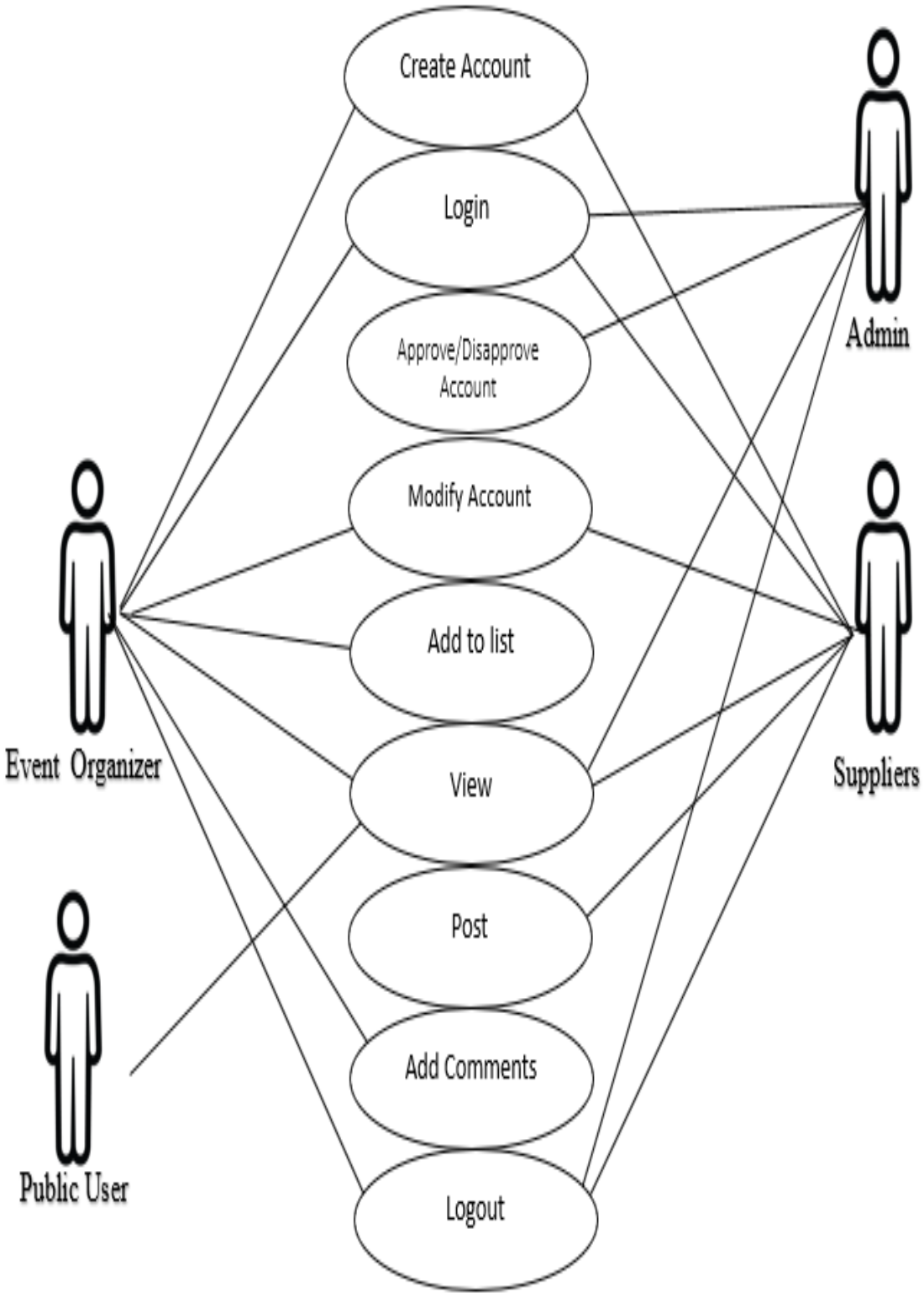



**Figure 8.0 Use Case Diagram**

Figure 8.0 Use Case Diagram shows the operations that the user can do in the Event Budget Manager. The event organizer can create an account, login, modify account, add to list a resource, view details, and logout. The suppliers can create accounts, login, modify their account, view details, post a resources package, and logout. The admin can login, approve/disapprove accounts, view, and logout. The public user can only view the posted packages and comments.

| Name | Add to List |
|---|---|
| Description | Event organizer click select to add a resource to the list. |
| Actor | Event Organizer |
| Preconditions | Login to the system. |
| Postconditions | Viewing the selected resources. |
| Main Course | 1. Login to the system.<br>2. Proceed to viewing the list of resources posted.<br>3. Select a supplier. |
| | 4. Click add to list to add the resources from the selected supplier.<br>5. View the selected resources. |
| Alternate Course | AC1: System encounters an error during selection.<br>1. Show warning message.<br>2. Return to main course step 2. |

**Table 1.0 Use Case Description (Select)**

| Name | Post |
|---|---|
| Description | Posting of resources package that can be seen the different event organizer. |
| Actor | Suppliers |
| Preconditions | Login to the system. |
| Postconditions | Viewing of posted resources. |
| Main Course | 1. Login to the system.<br>2. Click the add new resources.<br>3. Enter package details.<br>4. Click the post now button.<br>5. View the posted resources. |
| Alternate Course | AC1: System encounters an error during posting.<br>1. Show warning message.<br>2. Return to main course step 2. |

**Table 2.0 Use Case Description (Post)**

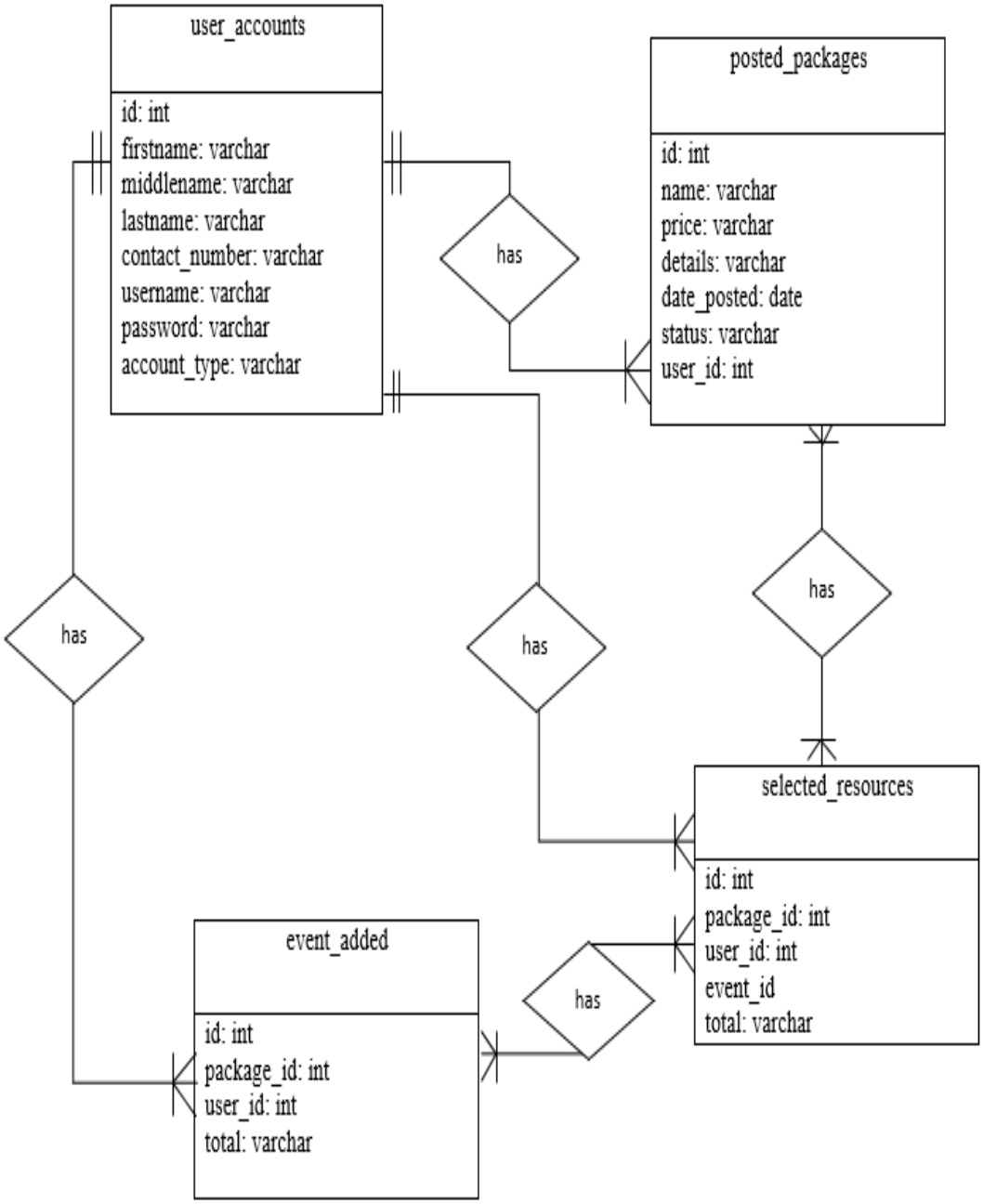



**Figure 9.0 Entity Relationship Diagram**

Figure 9.0 Entity Relationship Diagram shows the relationships of the entities in the database. There are three entities declared: user accounts, posted packages, selected resources, and event added. The entities user accounts and posted packages have a one-to-many cardinality, since one account can post many packages. The entities user accounts and selected resources have a one-to-many cardinality since a user account can have many selected resources. The entities user accounts and event added have a one-to-many cardinality, since user accounts can have many events added. The entities posted packages and selected resources have a many-to-many cardinality, since many packages can have many resources. The selected resources entities and event added can have a many-to-many cardinality, since many resources can have many event added.

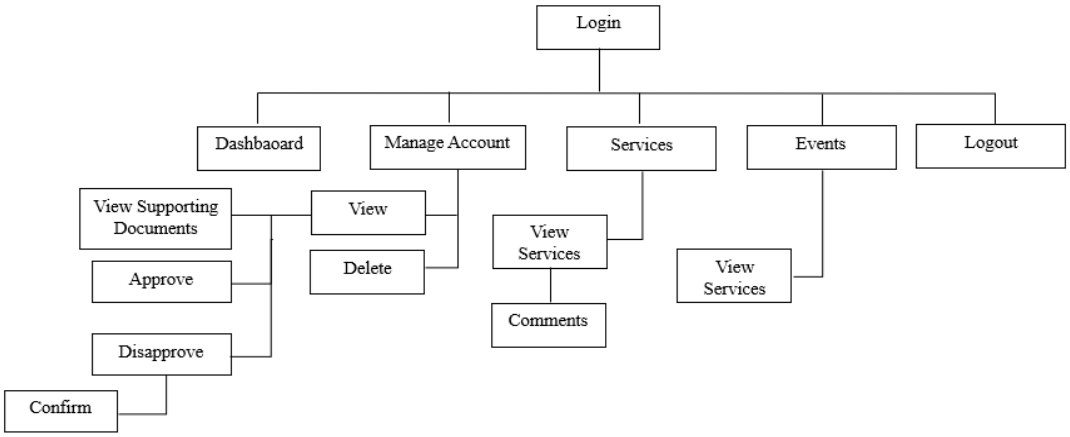



**Figure 10.0 Structured Diagram for Admin**

Figure 10.0 Structured Diagram for Admin shows the structure of the buttons and menus that the admin can use. The menus are dashboard, manage account, services, events, and logout. Under the manage account, approve, and disapprove buttons. The menu services have a view services button with comment buttons, and the events menu has only view resources.

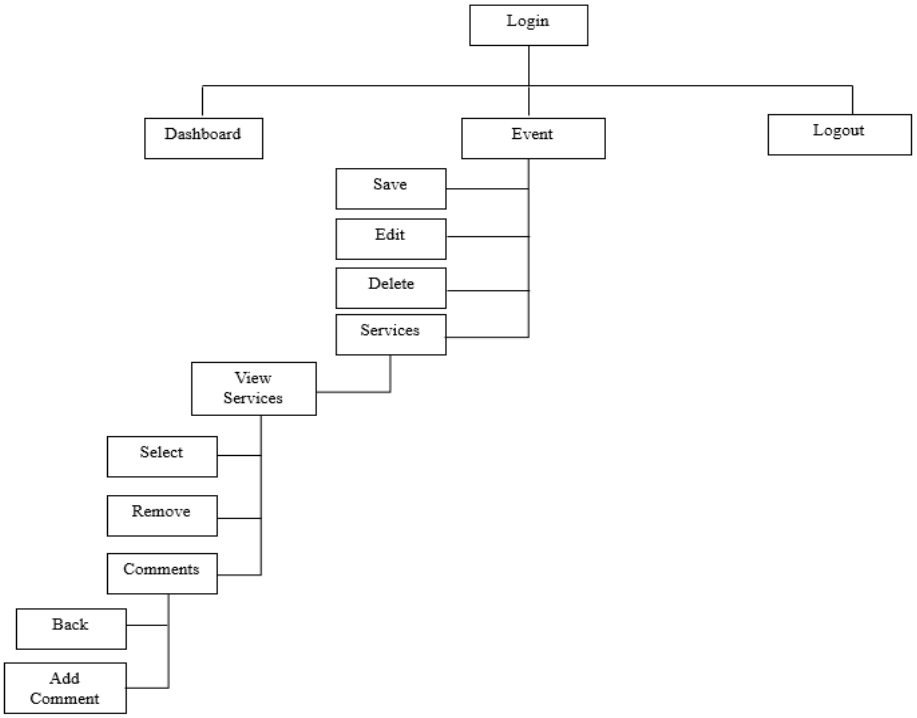



**Figure 11.0 Structured Diagram for Event Organizer**

Figure 11.0 Structured Diagram for Event Organizer shows the structure of the buttons and menus that the event organizer can use. The menus are dashboard, event, and logout. When the event organizer selects the event menu, there are buttons that will display: save, edit, delete, and services. When the services button is selected, a view services button will also appear. Once the user selects the view resources button, another set of buttons will display select, remove, and comments. On the comments button, an add comments button will also appear.

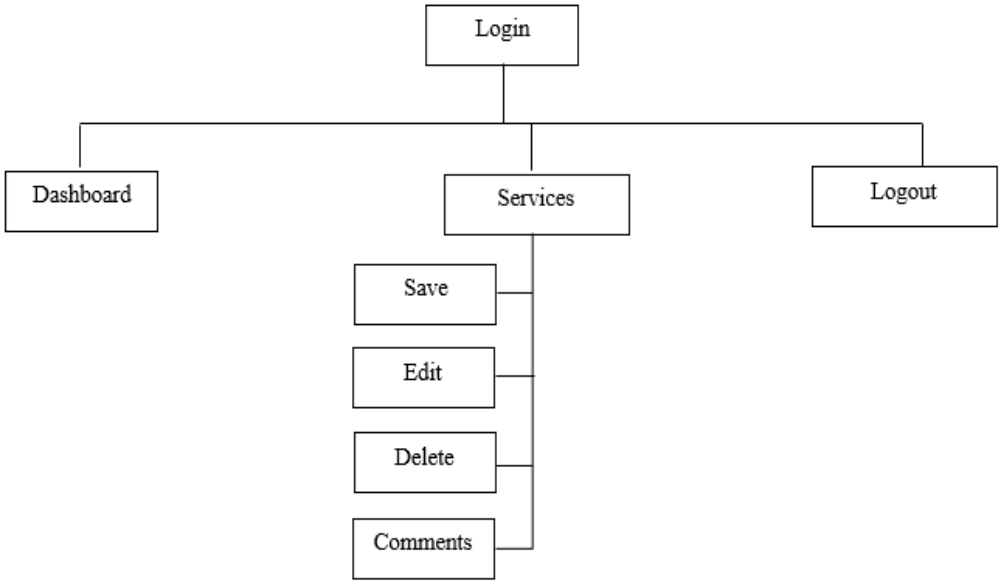



**Figure 12.0 Structured Diagram for Suppliers**

Figure 12.0 Structured Diagram for Suppliers shows the structure of the buttons and menus that the supplier can use: dashboard, services, and logout. Under the package menu, there will be buttons that will display, these are save, edit delete, and comments.

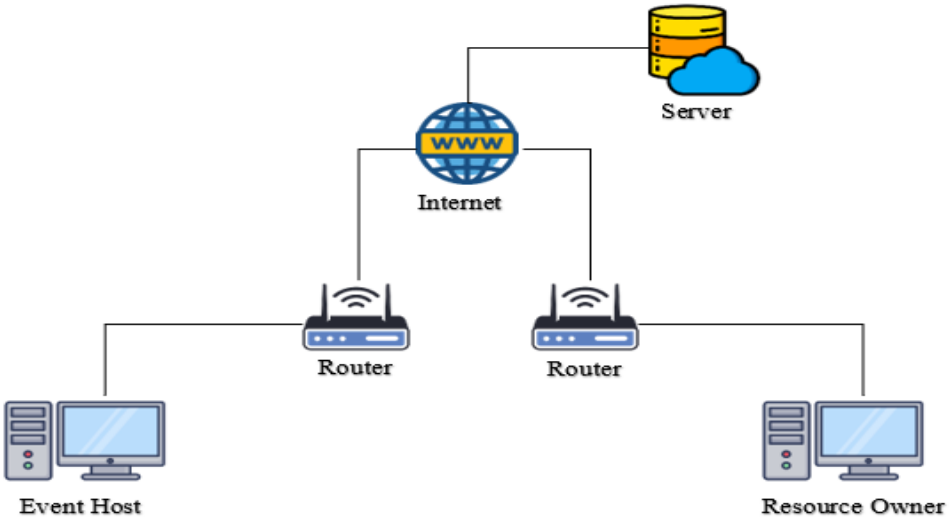



**Figure 13.0 System Architecture**

Figure 13.0 System Architecture shows the architecture of the system and its components. Event organizer and suppliers computers must be connected to the internet, the internet will connect both users through the routers and access the server and perform queries based on their request.

# CHAPTER IV

# DEVELOPMENT, TESTING AND IMPLEMENTATION

## Description and Prototype

The study "Event's Budget Manager" employs various methods in developing and implementing the system. This system aims to address the identified problem, providing enhanced user convenience.

| MONTH | July | | | | August | | | | September | | | | October | | | | November | | | | December | | | | January | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| WEEK | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 4 |
| Activities | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| DEFINE | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Planning | ■ | ■ | ■ | ■ | ■ | | | | | | | | | | | | | | | | | | | | | | | |
| Data Gathering | | | | | ■ | ■ | ■ | ■ | ■ | | | | | | | | | | | | | | | | | | | |
| Requirement Analysis | | | | | | | ■ | ■ | ■ | ■ | ■ | ■ | | | | | | | | | | | | | | | | |
| DESIGN | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| GUI | | | | | | | | | | | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | | | | | | | | |
| DEVELOP | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Coding | | | | | | | | | | | | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | | | | | | | |
| Testing | | | | | | | | | | | | | | | | | | | | | ■ | ■ | ■ | | | | | |
| DEPLOY | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Implementation | | | | | | | | | | | | | | | | | | | | | | | | | ■ | ■ | | |

**Table 3.0 Gantt Chart**

The Gantt chart presented in Table 3.0 outlines the timeline for developing the proposed study, commencing in July and concluding in January. The Define Phase, encompassing planning activities, spans the entirety of July and the initial week of August. Data gathering is allocated for the entire month of August and the initial week of September. Requirement analysis extends from the third week of August to the final week of September. Moving into the Design Phase, involving GUI design, the timeframe spans from the third week of September to the last week of November. Coding activities are slated from the fourth week of September to the initial week of December. The Deploy phase, which involves implementation, is scheduled for the first to second week of January.

## Technical Background

In todays generation, web applications have proven to be invaluable, offering enhanced functionalities that benefit various facet in different transactions, especially in businesses. These applications enable users to electronically store and manage information securely, thereby reducing the workloads of users and gaining recognition

from numerous beneficiaries. The functionality of web applications has made it very useful for every community that incorporate them.

Databases play an important role in efficient information management across different domains. The database provides a feature to the system, allowing it to store the information of supplier and event organizers. Once the supplier post a resources, it will be stored in the system database. The event organizer can access the information from the database since the system can retrieve stored information. The database plays a very important role in the system.

The significance of these aspects becomes evident in the context of the study. By leveraging the functionality of web applications, the study enables online query operations. The incorporation in the use of a database ensures that information obtained from different users is systematically organized and filtered for optimal utility.

**Development**

The system creation process employed by the proponents follows the 4D Methodology, guiding them through distinct phases to develop the project. One of its phases is development, wherein this methodology ensures a systematic approach, allowing the proponents to align the system's functionalities with user requirements and preferences. During the development, the proponent created the system design using HTML as the scripting language and JavaScript for some element movements. Additionally, in this phase, the proponent created the database of the system MySQL using XAMPP as the local server. After that, the proponent created the system's functions using the PHP programming language for the system's commands and queries.

**Program Specification**

Hardware Specification

- Laptop – core i3 processor
- RAM – 3GB

Software Specification

- Operating System - Windows 10
- Browser - Google Chrome
- XAMPP - Version v3.2.2

- Notepad ++ - v8.5.8
- PHP - Programming Language 8.2
- Database -MySQL v.8.1

**Module to Post Packages**

The codes below allow the supplier to post package to be seen by the event organizers. This line of code *<?php include('header.php'); ?>,* enable the system to include the codes of the header page, so that the design included will be applied on the page. The code *('../dbconnection.php');* enable the system to include the database connection from different file page, this will connect the system and the database. This line of codes *$name = $_POST['name']; $details = $_POST['details']; $pr.ice = $_POST['price']; ,* the values from the form being input by the supplier will be

```
<?php include('header.php'); ?>
<?php
// Database Connection file include ('../dbconnection.php');
if (isset($_POST['submit']))
{
// Getting the post values
$name = $_POST['name'];
$details = $_POST['details'];
$price = $_POST['price'];
$account_id = $_SESSION['id'];
$query = mysqli_query($conn, "insert into packages(name, details,
account_id, price) value('$name','$details', '$account_id','$price')
");
if
(!$query)
 {
echo "<script>alert('Something went wrong!');
</script>"; echo "<script type='text/javascript'> document.location =
'packages.php'; </script>";
} else
{ echo "<script>alert('New Package Added!');</script>"; } } //} ?>
```

**Module to Add Event**

This line of code *('../dbconnection.php')* enables the system to include the codes from the file in the dbconnection.php. It contains the codes that connect the database and the system. The codes *if (isset($_POST['submit']))* executes a function once the submit buttons is clicked by the user using the POST method. This line of code *$title = $_POST['title']; $event_date = $_POST['event_date']; $account_id = $_SESSION['id'];* runs to get the values of from the POST method from different variables declared on the code. The codes *]; $query = mysqli_query($conn, "insert*

*intoevent(title,event_date,account_id) value('$title','$event _date', '$account_id')");* executes to store the information from the variable in the system's database.

```
<?php
include ('../dbconnection.php'); if (isset($_POST['submit']))
{ // Getting the post values $title = $_POST['title'];
$event_date = $_POST['event_date'];
$account_id = $_SESSION['id'];
$query = mysqli_query($conn, "insert intoevent(title,event_date,
account_id) value('$title','$event _date', '$account_id')");if
(!$query) { echo"<script>ale rt('Something went wrong!');</script>";
 } else { echo "<script>alert('Event Saved!');</script>";
echo "<script type='text/javascript'> document.location ='event.php';
 </script>"; } } //} ?>
```

**Module to Add Packages**

This line of code *('../dbconnection.php')* enables the system to include the codes from the file in the dbconnection.php. It contains the codes that connect the database and the system. The codes *if (isset($_POST['submit']))* executes a function once the submit buttons is clicked by the user using the POST method. This line of code *$name = $_POST['name']; $details = $_POST['details']; $price = $_POST['price']; $account_id = $_SESSION['id'];* runs to get the values of from the POST method from different variables declared on the code.

```
<?php
('../dbconnection.php'); if (isset($_POST['submit']))
{
// Getting the post values
$name = $_POST['name'];
$details = $_POST['details'];
$price = $_POST['price'];
$account_id = $_SESSION['id'];
$query = mysqli_query($conn, "insert into packages
(name, details, account_id, price) value('$name','$details'
, '$account_id', '$price')");
if (!$query)
{
echo "<script>alert('Something went wrong!');</script>";
echo "<script type=' text/javascript'> document.location ='packages.php';
 </script>"; } else { echo "<script>alert('New Package Added!');
</script>"; } }
} ?>
```

**Module to Compute Total Budget**

The following codes provide a feature for the system to compute the total budget automatically. This line of code *includes('../dbconnection.php');* provides a connection to the system and the database. The codes *"SELECT SUM (packages.price) AS pack_price FROM event_resources INNER JOIN packages ON packages.id = event_*

*resources.packages_id WHERE event_id = '$id'");while($row=mysqli_fetch_array($query)){ $amo unt=$row['pack_price']; $formatted_amount = number _format($amount, 2, '.', ','); echo '₱' . $formatted_amo unt; // Output: ₱1,000.00} ?* selects the values or amount stored in the database for the selected packages by the event organizer, once selected, the system will get the SUM and display it to the system.

```
<?php
include('../dbconnection.php');
$id=$_GET ['id']; $query = mysqli_query($conn, "SELECT SUM
(packages.price AS pack_price FROM event_resources INNER
JOIN packages ON packages.id = event_ resources.packages_id
 WHERE event_id = '$id'");while($row=mysqli_fetch_array($query))
{ $amo unt=$row['pack_price']; $formatted_amount = number
_format($amount, 2, '.', ',');
echo '₱' . $formatted_amo unt; Output: ₱1,000.00}?>
```

**Load Testing**

The proponents tested the system to identify how it reacts when loaded with large amounts of data. The proponent tried to create an account and uploaded 50 valid IDs, and business permits with a picture size of 500kb. It saved 2-5 seconds. Then, tried to upload again with 600kb, and the system loaded up to 3-6 seconds. Therefore, the system loads slower once a bigger image size is uploaded.

**Stress Testing**

After the load testing, the proponents proceed to the stress testing. Stress testing helps the proponent determine when the system will encounter an error. The proponent tried to upload a Valid ID image with a size of 1mb, the system loaded for up to 15-30 seconds. Then attempted again to upload with a size of 1.5mb, but the system showed an error, stating that too much size of the image was being uploaded and the operation would be canceled. During the stress testing, it was found that the system cannot accept images with more than 1mb size. As the proponents found the problem, they added a code to the system, commanding to accept more than 1mb but not greater than 2mb.

**Beta Testing**

The proponent conducts system testing on the target users in the beta testing. The beta testing enables the proponents to collect feedback from the users. These feedbacks enable the proponent to improve the system, as the concerns will be addressed to improve its functionality. Beta testing is vital in enhancing system features and elements for a more effective system.



**Image 2.A. Beta Testing (Make-up Artist)**



**Image 2.B. Beta Testing (Supplier)**



**Image 2.C. Beta Testing (Photographer)**

The images above illustrate the documentation of beta testing conducted by the system's proponents and users. The proponents enabled a make-up artist named Richard Jainar in image 2.A, a resource owner, to utilize the system. Following Mr. Jainar, another user named Bacay shown in image 2.B was selected to test the system. These two users reside in Poblacion, Guipos, Zamboanga del Sur. Subsequently, the proponents traveled to a different location where they found another resource owner, Hilford Regidor, to use the system as shown in image 2.C.

Upon completion of beta testing, users provided feedback, with a majority expressing similar concept. They expressed gratitude for the developed system, which allows them to post their services and packages for viewing by multiple users, thereby increasing their potential customer base.

**Implementation Plan**

| Strategy | Activities | Person's Involve | Duration |
|---|---|---|---|
| Testing | Perform Load, Stress, and Beta Testing | Proponents, User | 3 Weeks |
| Approval from the users and panelists | Acceptance of the project | Proponents, Panelist | 1 Week and 1 Day |
| Installation | Installing the system together with its components | Proponents | 2 Days |
| Training | Train the users of the system | Proponents, Users | 2 Days |

**Table 4.0 Implementation Plan**

The Implementation Plan outlines the steps for introducing the system. In terms of strategy, the plan includes conducting load, stress, and beta testing over three weeks. Obtaining approval from project proponents and panelists is expected to take 1 week and 1 day. The installation phase involves setting up the system and its components within a two-days timeframe. Finally, two days is allocated for user training to ensure proficiency in using the system.

# CHAPTER V

# IMPLEMENTATION RESULT, DISCUSSION, CONCLUSION, AND RECOMMENDATION

## Implementation Result and Discussion

This chapter shows the flow and processes of the system, providing the discussions of the developed Event's Budget Manager. This involves screenshots of the system's user interface. The illustrations in this chapter provide information on how the system works based on the system's user interface.

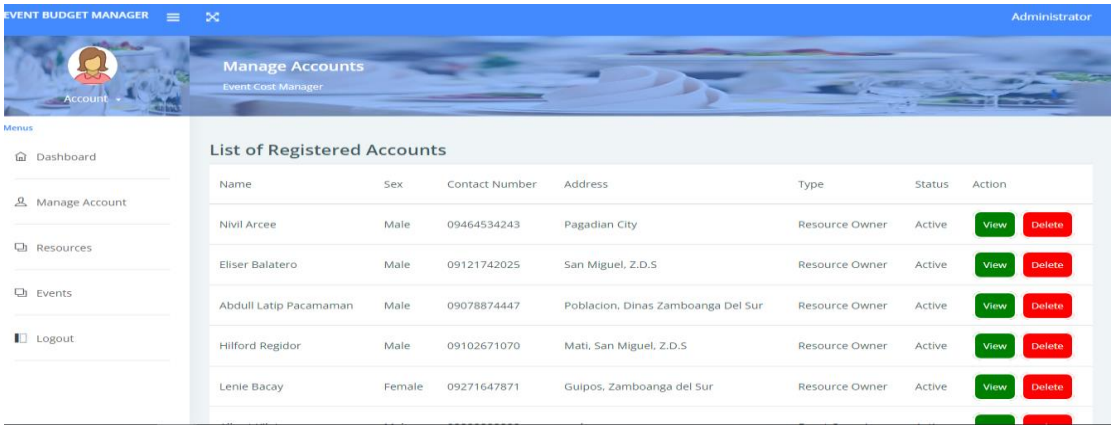



**Image 3.0 Manage Accounts (Admin)**

Image 3.0 displays the manage accounts interface accessible to the admin. Within this interface, the admin can view user accounts and approve or disapprove them based on the uploaded valid IDs or business permits. This process enhances the overall security of the system concerning user access.

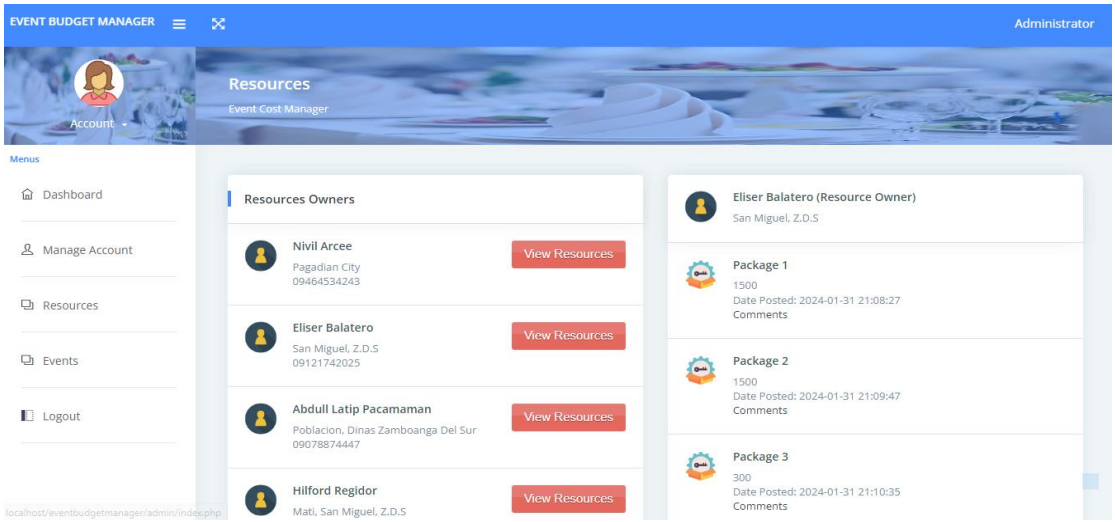



**Image 4 .0 View Resources Posted (Admin)**

Image 4.0 displays the resources posted on the system, accessed by the admin. The admin can monitor all the resources posted on the system, allowing for an

assessment of the credibility of the information posted. Additionally, the admin can review comments posted on resources which can be instrumental in the decision-making process.

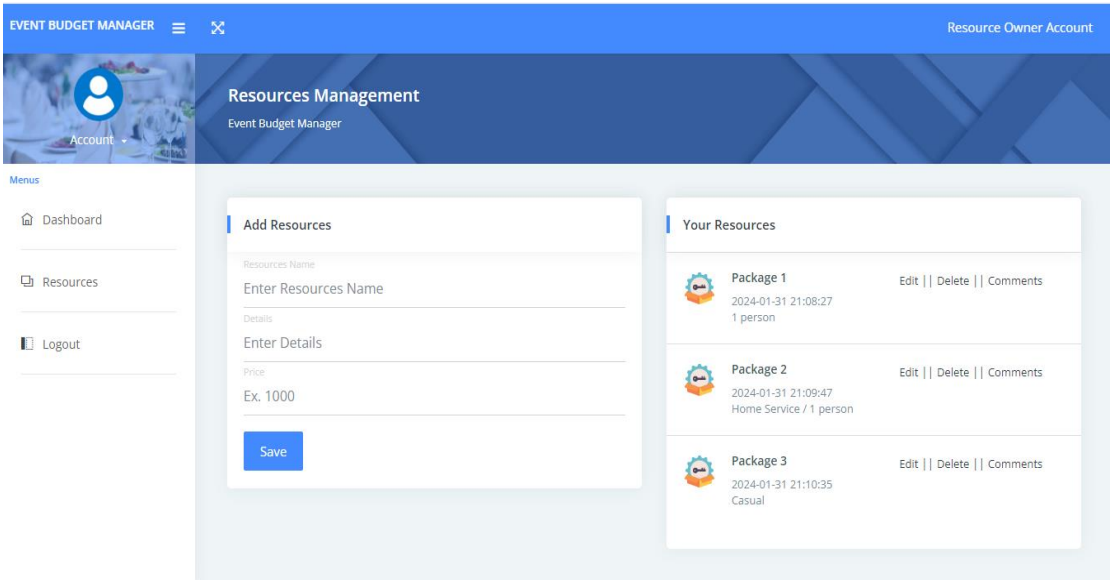



**Image 5.0 Manage Resources Posted (Supplier)**

Image 5.0 displays the interface accessible to suppliers for managing the posted resources. Within this interface, suppliers can input information about their resources, such as the name or package name, details, and pricing. The right section of the interface presents a list of resources already posted on the system. Here, the supplier can edit, delete, and review comments associated with each resource.

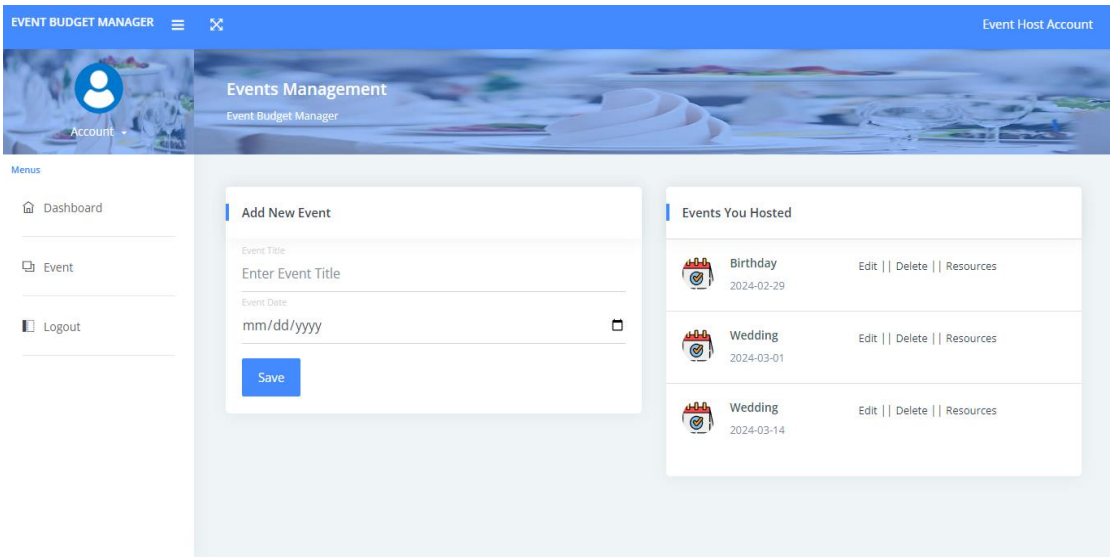



**Image 6.0 Management of Events (Event Organizer)**

Image 6.0 reveals the interface designed for event organizers to manage the list of hosted events. Within this interface, event organizers can enter the event title and its scheduled execution date. On the right section of the interface, organizers can make corrections to details, delete events, and oversee the management of required resources.

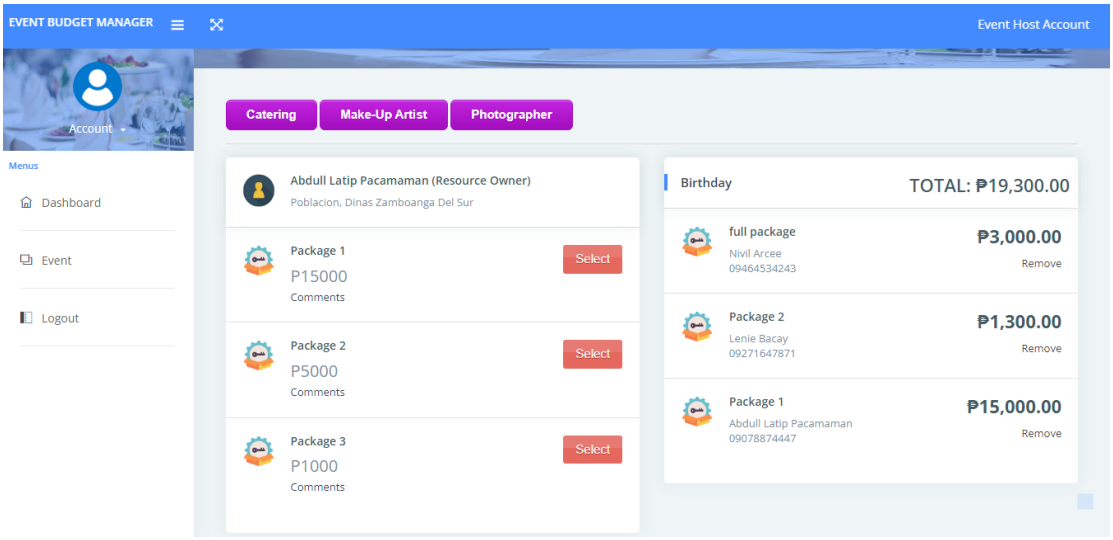



**Image 7.0 Selection of Resources (Event Organizer)**

Image 7.0 displays the interface accessible to the event organizer for managing the resources of a specific event. Within this interface, the event organizer can choose the required resources. Once a particular category is selected, the organizer can choose the package type they wish to avail. In the right section, the organizer can view the total amount needed for the hosted event, which is calculated based on the selected resources. This feature helps the event organizers to find and compute the needed resources for the event easily.

## Conclusion

In conclusion of this study, the developed Event Budget Manager was designed to allow users to have convenient, fast, and easy budgeting for the elements of an event. It provides a feature that allows event organizers to click and select needed resources for the specific event conveniently.

The developed system enables suppliers to post their resources and prices. This provides them an opportunity to attract more event organizers and increase profits. It is very useful for them since there will be a broader customer base that can see their resources.

The system has been tested, and its effectiveness is very useful based on the beta testing conducted. Users provided positive feedback regarding its effectiveness.

## Recommendation

For future researchers, the following are recommended:

- Implement a system feature allowing event organizers to reserve packages for specific resources. Simultaneously, provide a monitoring mechanism for supplier to track those who have availed of their resources.

- Integrate a notification feature into the system. Upon an event organizer availing specific resources, the system should generate and display notifications to inform all relevant parties.
- Enhance the study by incorporating sample pictures of the resources. This addition will serve as a visual reference for event organizers during the selection process, aiding them in making more informed choices.
- Add a system feature where the users can be notified once their registrations have been disapproved.

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